

Biotechnology English Jargon Quick Reference

Field-specific terms, contrast pairs, and high-pressure sentence frames

Audience: biotech scientists, translational researchers, assay-development teams, platform teams, program managers, alliance managers, business-development staff, and biotech executives

Focus: A high-level biotechnology English curriculum for platform science, assay design, translational evidence, biomarkers, IP, partnerships, investor updates, regulatory-adjacent planning, and scientific-business dialogue.

Designed for advanced ESL learners who already use professional English and need industry-specific terminology, realistic meetings, role-play pressure, careful pushback, and polished workplace outputs.

Teaching stance: this is language and workplace-communication training, not legal, medical, financial, safety, or regulatory advice. Instructors should connect every scenario to the learner's current company policies, local rules, and approved procedures.

Nomenclature and Jargon

These are classroom working definitions. Learners should adapt wording to their organization's policies, systems, and local regulatory environment.

Platform Technology and Scientific Thesis

Term	Working meaning	Collocations	Contrast and workplace line
platform	In biotechnology, platform is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to explain platform promise without overstating translation to products.	clarify platform; document platform	Contrast: platform should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify platform and the evidence that supports it.
modality	In biotechnology, modality is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to explain platform promise without overstating translation to products.	clarify modality; document modality	Contrast: modality should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify modality and the evidence that supports it.
proof of concept	In biotechnology, proof of concept is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to explain platform promise without overstating translation to products.	clarify proof of concept; document proof of concept	Contrast: proof of concept should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify proof of concept and the evidence that supports it.
translatability	Likelihood that a finding in one model, setting, or population will predict results in the intended real-world or clinical setting.	clarify translatability; document translatability	Contrast: translatability should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify translatability and the evidence that supports it.

Assay Development and Reproducibility

Term	Working meaning	Collocations	Contrast and workplace line
assay	Laboratory procedure used to measure the presence, amount, activity, or quality of a biological target.	validate the assay; run the assay	Contrast: An assay is a defined laboratory measurement method, not proof that a result will translate to clinical benefit. Example: The team will validate the assay's precision before using it for a development decision.
sensitivity	Ability of a test, model, or process to correctly identify true positives or detect a change in an input.	clarify sensitivity; document sensitivity	Contrast: sensitivity should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify sensitivity and the evidence that supports it.
specificity	Ability of a test, model, or process to correctly identify true negatives and avoid false positives.	clarify specificity; document specificity	Contrast: specificity should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify specificity and the evidence that supports it.

Term	Working meaning	Collocations	Contrast and workplace line
reproducibility	In biotechnology, reproducibility is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to discuss assay performance, variability, and limitations.	clarify reproducibility; document reproducibility	Contrast: reproducibility should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify reproducibility and the evidence that supports it.

Biomarkers and Translational Strategy

Term	Working meaning	Collocations	Contrast and workplace line
biomarker	Measurable biological characteristic used to indicate a condition, response, exposure, or disease-related process.	validate the biomarker; stratify by biomarker	Contrast: A biomarker can indicate a biological state or response; it is not automatically a clinically useful endpoint. Example: The protocol will stratify patients by biomarker status only after the assay is validated.
patient stratification	Grouping patients by characteristics that may affect disease course, treatment response, safety, or trial analysis.	clarify patient stratification; document patient stratification	Contrast: patient stratification should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify patient stratification and the evidence that supports it.
surrogate endpoint	Measure expected to predict clinical benefit but not itself a direct measure of how a patient feels, functions, or survives.	clarify surrogate endpoint; document surrogate endpoint	Contrast: surrogate endpoint should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify surrogate endpoint and the evidence that supports it.
validation	In biotechnology, validation is a repeatable control mechanism used to prevent, detect, or confirm conditions before work proceeds. Use it when teams need to connect preclinical signals to patient selection and clinical hypotheses.	clarify validation; document validation	Contrast: validation should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify validation and the evidence that supports it.

Preclinical Data Packages

Term	Working meaning	Collocations	Contrast and workplace line
in vitro	Performed outside a living organism, typically in a laboratory vessel, cell system, or controlled assay.	clarify in vitro; document in vitro	Contrast: in vitro should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify in vitro and the evidence that supports it.
in vivo	Performed in a living organism, such as an animal or human study participant.	clarify in vivo; document in vivo	Contrast: in vivo should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify in vivo and the evidence that supports it.

Term	Working meaning	Collocations	Contrast and workplace line
toxicity	In biotechnology, toxicity is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to describe animal, in vitro, and tox findings with appropriate caution.	clarify toxicity; document toxicity	Contrast: toxicity should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify toxicity and the evidence that supports it.
safety margin	In biotechnology, safety margin is a measurable indicator used to quantify performance, exposure, capacity, quality, or change over time. Use it when teams need to describe animal, in vitro, and tox findings with appropriate caution.	track safety margin; explain a change in safety margin	Contrast: safety margin measures a result or condition; it does not, by itself, prove the cause. Example: Before we act, let us track safety margin by segment and explain what changed.

CMC and Scale-Up in Biotech

Term	Working meaning	Collocations	Contrast and workplace line
CMC	Chemistry, manufacturing, and controls information that shows how a drug substance or product is made and consistently controlled.	clarify CMC; document CMC	Contrast: CMC should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify CMC and the evidence that supports it.
yield	In biotechnology, yield is a measurable indicator used to quantify performance, exposure, capacity, quality, or change over time. Use it when teams need to explain manufacturing feasibility before process maturity is complete.	improve yield; analyze yield loss	Contrast: Yield measures conforming output, not the root cause of a process change. Example: The fab is analyzing yield loss by tool, layer, and time window.
comparability	In biotechnology, comparability is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to explain manufacturing feasibility before process maturity is complete.	clarify comparability; document comparability	Contrast: comparability should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify comparability and the evidence that supports it.
tech transfer	Controlled transfer of product, process, method, or manufacturing knowledge between teams, sites, or organizations.	clarify tech transfer; document tech transfer	Contrast: tech transfer should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify tech transfer and the evidence that supports it.

IP, Freedom to Operate, and Collaboration

Term	Working meaning	Collocations	Contrast and workplace line
IP	In biotechnology, IP is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to use careful language around patents and partner boundaries.	clarify IP; document IP	Contrast: IP should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify IP and the evidence that supports it.
freedom to operate	Assessment of whether a product or activity can proceed without infringing another party's enforceable intellectual-property rights.	track freedom to operate; explain a change in freedom to operate	Contrast: freedom to operate measures a result or condition; it does not, by itself, prove the cause. Example: Before we act, let us track freedom to operate by segment and explain what changed.

Term	Working meaning	Collocations	Contrast and workplace line
material transfer agreement	In biotechnology, material transfer agreement is a negotiated legal or commercial arrangement that allocates rights, obligations, remedies, or risk. Use it when teams need to use careful language around patents and partner boundaries.	clarify material transfer agreement; document material transfer agreement	Contrast: material transfer agreement should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify material transfer agreement and the evidence that supports it.
publication rights	In biotechnology, publication rights is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to use careful language around patents and partner boundaries.	clarify publication rights; document publication rights	Contrast: publication rights should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify publication rights and the evidence that supports it.

Investor and Board Updates

Term	Working meaning	Collocations	Contrast and workplace line
milestone	In biotechnology, milestone is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to translate science into milestone, runway, risk, and decision language.	clarify milestone; document milestone	Contrast: milestone should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify milestone and the evidence that supports it.
runway	In biotechnology, runway is a measurable indicator used to quantify performance, exposure, capacity, quality, or change over time. Use it when teams need to translate science into milestone, runway, risk, and decision language.	track runway; explain a change in runway	Contrast: runway measures a result or condition; it does not, by itself, prove the cause. Example: Before we act, let us track runway by segment and explain what changed.
inflection point	In biotechnology, inflection point is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to translate science into milestone, runway, risk, and decision language.	clarify inflection point; document inflection point	Contrast: inflection point should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify inflection point and the evidence that supports it.
risk mitigation	In biotechnology, risk mitigation is a condition or event that can create harm, loss, noncompliance, delay, or unacceptable performance. Use it when teams need to translate science into milestone, runway, risk, and decision language.	assess risk mitigation; mitigate risk mitigation	Contrast: risk mitigation identifies a possible or current exposure; it is not the same as accepting that exposure. Example: We need to assess risk mitigation, name the owner, and agree on the trigger for escalation.

Partnering and Business Development

Term	Working meaning	Collocations	Contrast and workplace line
term sheet	In biotechnology, term sheet is a negotiated legal or commercial arrangement that allocates rights, obligations, remedies, or risk. Use it when teams need to discuss deal value without losing scientific caveats.	clarify term sheet; document term sheet	Contrast: term sheet should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify term sheet and the evidence that supports it.

Term	Working meaning	Collocations	Contrast and workplace line
option	In biotechnology, option is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to discuss deal value without losing scientific caveats.	clarify option; document option	Contrast: option should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify option and the evidence that supports it.
exclusivity	In biotechnology, exclusivity is a field-specific concept used to locate the relevant fact, boundary, or decision variable. Use it when teams need to discuss deal value without losing scientific caveats.	clarify exclusivity; document exclusivity	Contrast: exclusivity should be defined against the relevant evidence and decision boundary, rather than used as a vague label. Example: Before we proceed, please clarify exclusivity and the evidence that supports it.
due diligence	Structured review of facts, risks, obligations, and evidence before a transaction, onboarding, investment, or decision.	conduct due diligence; complete due diligence	Contrast: Due diligence tests facts and risks before commitment; it is not a ceremonial step after the decision is made. Example: The team will complete due diligence on data practices before onboarding the vendor.

Industry-Specific Meeting Moves

Situation	Useful language
Platform Technology and Scientific Thesis	Before we commit, I want to confirm platform, modality, the owner, and the evidence behind the decision. If the evidence supports specific models and indications, not every future use case., I recommend we document the risk and agree on the next step.
Assay Development and Reproducibility	Before we commit, I want to confirm assay, sensitivity, the owner, and the evidence behind the decision. If precision, sensitivity, specificity, controls, and reproducibility need review., I recommend we document the risk and agree on the next step.
Biomarkers and Translational Strategy	Before we commit, I want to confirm biomarker, patient stratification, the owner, and the evidence behind the decision. If biological plausibility, patient relevance, and validation status are different., I recommend we document the risk and agree on the next step.
Preclinical Data Packages	Before we commit, I want to confirm in vitro, in vivo, the owner, and the evidence behind the decision. If model relevance, dose, exposure, safety margin, and limitations must be named., I recommend we document the risk and agree on the next step.
CMC and Scale-Up in Biotech	Before we commit, I want to confirm CMC, yield, the owner, and the evidence behind the decision. If yield, process control, comparability, and supply-chain risk are unresolved., I recommend we document the risk and agree on the next step.
IP, Freedom to Operate, and Collaboration	Before we commit, I want to confirm IP, freedom to operate, the owner, and the evidence behind the decision. If confidentiality, background ip, publication rights, and freedom to operate matter., I recommend we document the risk and agree on the next step.
Investor and Board Updates	Before we commit, I want to confirm milestone, runway, the owner, and the evidence behind the decision. If credibility requires clear milestones, risks, assumptions, and mitigation., I recommend we document the risk and agree on the next step.
Partnering and Business Development	Before we commit, I want to confirm term sheet, option, the owner, and the evidence behind the decision. If diligence, territory, field, economics, governance, and data rights need definition., I recommend we document the risk and agree on the next step.

High-pressure pushback frames

- I understand the urgency. The risk is that we move faster than the evidence or process supports.
- I am not blocking the goal. I am naming the condition we need before the decision is safe and credible.

- If we accept this risk, we should name the owner, document the assumption, and define the trigger for escalation.
- That may be possible, but not under the current scope, timeline, or approval path.
- Let's separate what we know, what we assume, and what still needs confirmation.

Contrast Pairs

Do not confuse	Useful distinction
platform vs translatability	In platform technology and scientific thesis, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.
assay vs reproducibility	In assay development and reproducibility, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.
biomarker vs validation	In biomarkers and translational strategy, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.
in vitro vs safety margin	In preclinical data packages, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.
CMC vs tech transfer	In cmc and scale-up in biotech, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.
IP vs publication rights	In ip, freedom to operate, and collaboration, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.
milestone vs risk mitigation	In investor and board updates, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.
term sheet vs due diligence	In partnering and business development, define whether the discussion is about the current fact pattern, the controlling process, the stakeholder pressure, or the final decision.